



Re: 5/28/2026—RTSL UW CVH Modeling Research Notes

From Zina Abourjeily <zabourjeily@resolvetosavelives.org>

Date Wed 7/8/2026 11:40 AM

To William R Garcia <wrgg@uw.edu>

Hi William,

Here you go!

discretionary (salt added at table and during cooking)	packaged food (salty seasonings, home, other)	restaurant and street	public food procurement	inherent in foods
23.1%	41.5%	20.5%	5.0%	10.0%

Thanks! Looking forward to learning about the preliminary results tomorrow.

Very Best,
Zina

From: William R Garcia <wrgg@uw.edu>

Date: Wednesday, July 8, 2026 at 12:37 PM

To: Zina Abourjeily <zabourjeily@resolvetosavelives.org>

Subject: Re: 5/28/2026—RTSL UW CVH Modeling Research Notes

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Hi Zina,

Yes. I'm compiling the preliminary modeling results for tomorrow's meeting. Currently, Nigeria is listed as Blank (serving as a proxy for Cameroon). For completeness, it would be helpful to have at least an estimate, while noting that everything remains preliminary.

Best,

William

From: Zina Abourjeily <zabourjeily@resolvetosavelives.org>
Sent: Wednesday, July 8, 2026 11:17 AM
To: William R Garcia <wrgg@uw.edu>
Subject: Re: 5/28/2026—RTSL UW CVH Modeling Research Notes

Hi William,

Not yet...still working with Matti to figure that out. Would estimates still be helpful in the meantime, knowing that they will change?

Very best,
Zina

From: William R Garcia <wrgg@uw.edu>
Date: Wednesday, July 8, 2026 at 12:16 PM
To: Zina Abourjeily <zabourjeily@resolvetosavelives.org>
Subject: Re: 5/28/2026—RTSL UW CVH Modeling Research Notes

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Thanks, Zina!

Did you happen to receive the data from Nigeria on the proportion of sodium intake by source?

Best,

William

From: Zina Abourjeily <zabourjeily@resolvetosavelives.org>
Sent: Monday, July 6, 2026 8:01 AM
To: William R Garcia <wrgg@uw.edu>; David Watkins <davidaw@uw.edu>; mmarklu1 <mmarklu1@jhu.edu>
Cc: Andrew Moran <amoran@resolvetosavelives.org>; Laura Cobb <lcobb@resolvetosavelives.org>; Nicole Ide <nide@resolvetosavelives.org>
Subject: Re: 5/28/2026—RTSL UW CVH Modeling Research Notes

Hi William,

Happy Monday!

The priority countries for now are what are listed in the table titled "Summary table of sources." Only 10 for now.

Very Best,
Zina

From: William R Garcia <wrgg@uw.edu>
Date: Thursday, July 2, 2026 at 5:54 PM
To: Zina Abourjeily <zabourjeily@resolvetosavelives.org>; davidaw <davidaw@uw.edu>; mmarklu1 <mmarklu1@jhu.edu>

Cc: Andrew Moran <amoran@resolvetosavelives.org>; Laura Cobb <lcobb@resolvetosavelives.org>; Nicole Ide <nide@resolvetosavelives.org>
Subject: Re: 5/28/2026—RTSL UW CVH Modeling Research Notes

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Hi Zina,

Thank you for the update. We will use the estimates for sodium from discretionary, packaged food, and public food procurement sources as suggested.

Could you also please share the list of the 64(??) prioritized countries we discussed last time?

Best,

William

From: Zina Abourjeily <zabourjeily@resolvetosavelives.org>

Sent: Friday, June 26, 2026 1:06 PM

To: William R Garcia <wrgg@uw.edu>; David Watkins <davidaw@uw.edu>; mmarklu1 <mmarklu1@jhu.edu>

Cc: Andrew Moran <amoran@resolvetosavelives.org>; Laura Cobb <lcobb@resolvetosavelives.org>; Nicole Ide <nide@resolvetosavelives.org>

Subject: Re: 5/28/2026—RTSL UW CVH Modeling Research Notes

Hi William,

The document previously shared with you for the effect sizes ([📄 Sodium Lives Saved Calculator_19May2026.docx](#)) has been updated to have the relevant data available on the sources of sodium, with the exception of Nigeria and Cameroon. We may still make some changes to the exact numbers, but will be sure to let you know when/if we do so.

Please use the estimates for sodium from discretionary, packaged food, and public food procurement sources. Let me know if you have any questions!

Matti ([@mmarklu1](#))—**could you help us determine the breakdown for Nigeria of discretionary, packaged food, and outside the home sources of sodium**, with the assumption that bouillon and condiments are considered packaged food? If you could share if/what is possible, based on the data you have access to, that would be helpful!

Additionally, could you look at the Philippines estimates (pasted below for your reference. Does this seem plausible to you based on the extensive amount of work you've done in the Philippines?

discretionary	packaged food	restaurant and street	public food procurement	inherent in foods	Total
32.0%	43.0%	10.0%	5.0%	10.0%	100.0%

Very Best,
Zina

From: Zina Abourjeily <zabourjeily@resolvetosavelives.org>

Date: Thursday, May 28, 2026 at 5:15 PM

To: Andrew Moran <amoran@resolvetosavelives.org>; Laura Cobb <lcobb@resolvetosavelives.org>; Nicole Ide <nide@resolvetosavelives.org>; mmarklu1 <mmarklu1@jhu.edu>; William R Garcia <wrgg@uw.edu>; davidaw <davidaw@uw.edu>

Subject: 5/28/2026—RTSL UW CVH Modeling Research Notes

Hi all,

Thank you for a productive call! Pasted below are notes from today. The first half are directly what appeared in William's slides.

5/28/2026—RTSL UW CVH Modeling Research

Attendees:

- UW: William, David
- RTSL: Laura, Nicole, Zina, Andrew (partial)
- JHU: Matti

Notes:

Presentation from William:

Project Objective and Policy Impact

- Estimate national-scale health impacts of dietary sodium reduction policies using a population CVD model.
- Quantify deaths and cases of ischemic heart disease (IHD), stroke, and hypertensive heart disease, as well as DALYs averted over 2026–2050.
- Conduct a full economic benefit–cost analysis to support government decision-making.
- Focus on RTSL partner countries and on policies that governments can realistically implement.

Where We Left Off

- **Modeling framework:** Multi-state CVD life-table (Well → Sick → Dead) simulated annually over 2026–2050 for ages 20–95, four CVD causes (IHD, stroke, HHD, other CVD), by age × sex × country.
- **Outcome metrics:** Deaths averted, IHD/stroke/HHD cases averted, DALYs averted, plus full benefit–cost analysis (NPV, BCR).
- **Baseline scenario design:** Five SHAKE-aligned policy bundles with total sodium-reduction efficacy salt_eff (η) ranging from 0 (baseline) to 0.250 (full mandatory package).
- **Causal pathway:** Sodium → SBP (Filippini dose–response) → CVD incidence (Ettehad per-10-mmHg RRs) → life-table transitions.
- **Deliverables:** Peer-reviewed paper + RTSL report + interactive Shiny tool, originally targeted for end of June 2026 (per Apr 24 timeline).

Sources of Sodium — Current State

Zina and Nicole have been compiling literature on sodium shares across three categories:

- **Packaged foods:** Processed and ultra-processed sources, including salt-based condiments (fish sauce, soy sauce) where separately reported.
- **Discretionary sources:** Salt or MSG added while cooking at home or at the table.

- **Outside-the-home:** Restaurants, cafeterias/canteens, street food. Approximately 1/3 of outside-home assumed to be institutional (public food procurement).

This compilation feeds directly into the **Sodium Lives Saved Calculator** (next section) and determines how intervention effects are attributed to the correct sodium source.

Key method: Where country-specific data are unavailable, discretionary share is predicted from GDP per capita using the Bhat/Marklund regression (Bhat et al. 2020). Processed share is predicted from discretionary share. Floors: 10% discretionary (USA), 2% packaged (Ethiopia).

Calculator Overview

- **Purpose:** Estimate national-scale lives saved from a portfolio of sodium-reducing interventions.
- **Six intervention areas:** (A) Public food procurement, (B) Behavior change / mass media, (C) Low-sodium salt substitutes, (D) Front-of-pack labeling, (E) Mandatory salt targets, (F) Fiscal policy / taxation.
- **Source disaggregation:** Sodium intake split into discretionary, packaged, and outside-the-home (with ~1/3 of outside-home = institutional). Each intervention acts on its target source.
- **Multiplicative combination:** Effects within a source are combined multiplicatively, not additively, to prevent >100% reductions.
E.g., tax (13.7%) + FOPL (15%) on packaged $\Rightarrow 1 - (1 - 0.137)(1 - 0.15) = 26.6\%$.
- **Discretionary share:** Predicted from GDP per capita (Bhat et al. 2020); floors at 10% (USA) and 2% packaged (Ethiopia).
- **Packaged-food growth:** Projected using Vandevijvere et al. 2019 regional growth rates (Vandevijvere et al. 2019), capped at Singapore (4.3 g Na/day from packaged).
- **Demo:** https://dcp-uw.shinyapps.io/salt_demo/
Code: <https://github.com/Disease-Control-Priorities/sodium>

Intervention	Source targeted	Study	Δ mg/day	Δ (%)	Time horizon
A. Public food procure- ment	Institutional	Gase 2011	−233	7%	Modeled
B. Behavior change / me- dia	Discret. + Packaged	Do 2016	—	5%	1 year
C. LSSS	Discretionary	Huang 2026	—	TBD	TBD
D. Front-of-pack labeling	Packaged	Taillie 2024	−73.3	13.8%	3 years
E. Mandatory salt targets	Packaged	Strauss-Kruger 2022	−480	15.2%	4.56 years
F. Fiscal policy / taxation	Packaged (food cat- egories)	Saxena 2025	−100	10%	20 years

Discussion

Behavior change

- Updated the estimate to be based on the UK; however, this component may ultimately not be used.

Fiscal policies

- UW raised concerns about the political feasibility of food taxes, particularly taxes on food categories vs. salt.
 - RTSL is not interested in salt taxes, given salt’s low cost.
- UW questioned the generalizability of the Saxena estimate, as it is country-specific.

- o Matti noted that estimates should not differ substantially from others that are also based on
- UW also raised concerns about using modeled fiscal policy effect sizes for empirical scrutiny.
 - o RTSL is less focused on publishing a paper and more focused on **BP reinvestment proposal**

Country scope

- Current list includes 29 countries where RTSL conducts sodium work; PAHO adds significantly this count.
- The list may be refined based on countries we are more involved in
- Analysis could also highlight gaps and serve as a call for improved data collection, but this is less RTSL's concern as it relates to reinvestment

Sources of Sodium

- LSS study:
 - o Primary source: Bhat et al., with a method to select among multiple data sources.
 - Found that sodium sources are associated with level of economic development; conducted a regression using GDP.
 - Non-discretionary sodium includes sodium intrinsic to foods.
 - The analysis did not account for changes in sodium sources over time.
- Laura noted that drivers extend beyond GDP (e.g., high condiment or bouillon use in countries such as Vietnam and Nigeria).
- Data are highly time-sensitive; older data may be outdated.

Proposed approach

- Start with available data, then apply GDP-based mapping where data are missing (combined approach).
- Consider incorporating uncertainty into lives saved estimates as a potential call for better data (though RTSL is less supportive of this approach due to past challenges estimating sodium sources).

Policy framing

- The policy goal is not always to maximize lives saved in modeled interventions.
 - o Example: although Ethiopia has low packaged food consumption, packaged food policies are still prioritized for prevention.
 - o A counterfactual scenario may better capture this preventative framing.
- Do not map to SHAKE; this is no longer how the new SHAKE discussed the interventions

LSSS

- Include potassium supplementation.
- Focus on scenarios 2, 4, and 5.

Timeline

- September 2026 for reinvestment deliverables.

Expected Output from UW

- Memo including tables and figures (content to be finalized).

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